

Assignment Report

Title: **Compiler Principles**

The Ninth Theory Homework

Name	林宏宇
Student ID	23320093
School	计算机学院
Major	计算机科学与技术
Instructor	李文军

[ABSTRACT]

After learning the lecture 9 of compiler principles[1], we got the homework at the end of the lecture.

There are some requirements for the homework:

- The file should be named as **“学号 _ 姓名 _ 第 i 次理论作业”**, where i is the number of the homework.
- The file should be in docx or **pdf** format.
- All the homework should be sent to the TA's email: **406648896@qq.com**
- The deadline for the homework is **24:00 on Sunday of the week** when the homework is assigned.

Keywords: Compiler Principles

Contents

1 Exercise 9 1
1.1 Exercise 9.1 1
1.2 Exercise 9.2 1
1.3 Exercise 9.3 2

References 4

1 Exercise 9

1.1 Exercise 9.1

1.1.1 Problem

Let A be declared as a $[2..4] \times [1..5]$ array of integers, and each integer occupies 4 bytes. What is the translation result of input token string: $x := A[3, 2]$?

- Tips: use the translation scheme for Pascal.

1.1.2 Solution

For Pascal arrays, the row-major offset of $A[3, 2]$ is

$$(3 - 2) \times 5 + (2 - 1) = 6.$$

Since each integer occupies 4 bytes, the byte offset is $6 \times 4 = 24$. Thus the generated translation is

$$\begin{aligned} t_1 &= 3 - 2 \\ t_2 &= t_1 \times 5 \\ t_3 &= 2 - 1 \\ t_4 &= t_2 + t_3 \\ t_5 &= 4 \times t_4 \\ t_6 &= base_A + t_5 \\ x &= t_6 \end{aligned}$$

1.2 Exercise 9.2

1.2.1 Problem

Let a be declared as a 5×6 array of integers, and each integer occupies 4 bytes. What is the translation result of input token string: $i = a[3][2]$?

- Tips: use the translation scheme for C/C++.

1.2.2 Solution

In C/C++, arrays are stored in row-major order. The byte offset of $a[3][2]$ is

$$(3 \times 6 + 2) \times 4 = 80.$$

So the translation result is

$$t_1 = 3 \times 6$$
$$t_2 = t_1 + 2$$
$$t_3 = 4 \times t_2$$
$$t_4 = a[t_3]$$
$$i = t_4$$

1.3 Exercise 9.3

1.3.1 Problem

What is the translation result of input token string: `x < 100 || x > 200 && x != y`?

- Tips: use the translation scheme for boolean expressions with backpatching.
- Suppose that the start position of the generated code is 100.

1.3.2 Solution

Using short-circuit evaluation and backpatching, one valid generated code sequence is

$$
\begin{aligned}
&100: \text{ if } x < 100 \text{ goto } _ \\
&101: \text{ goto } 102 \\
&102: \text{ if } x > 200 \text{ goto } 104 \\
&103: \text{ goto } _ \\
&104: \text{ if } x \neq y \text{ goto } _ \\
&105: \text{ goto } _
\end{aligned}
$$

The blank targets are filled by backpatching after the true and false lists of the whole expression are known.

References

[1] JUN LI W. Lecture09. principles of compiler construction[Z]. 2026.